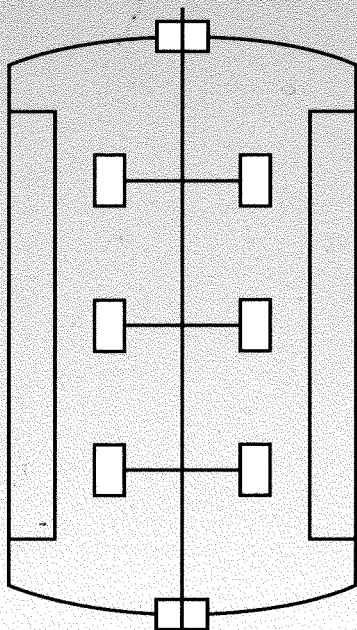
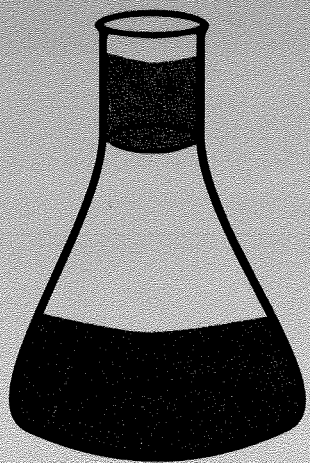
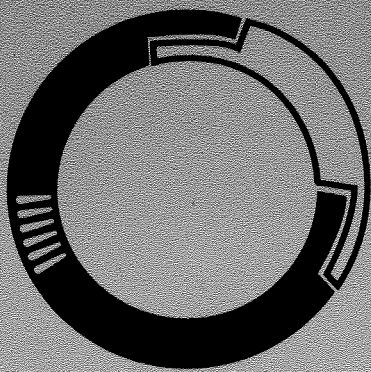




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Principles of Fermentation Technology

PETER F. STANBURY
B.Sc., M.Sc., D.I.C.

*Division of Biosciences,
University of Hertfordshire,
Hatfield, U.K.*

ALLAN WHITAKER
M.Sc., Ph.D., A.R.C.S., D.I.C.

*Division of Biosciences,
University of Hertfordshire*

STEPHEN J. HALL
B.Sc., M.Sc., Ph.D.

*Division of Chemical Sciences,
University of Hertfordshire*

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CHAPTER 1

An Introduction to Fermentation Processes

THE TERM 'fermentation' is derived from the Latin verb *fervere*, to boil, thus describing the appearance of the action of yeast on extracts of fruit or malted grain. The boiling appearance is due to the production of carbon dioxide bubbles caused by the anaerobic catabolism of the sugars present in the extract. However, fermentation has come to have different meanings to biochemists and to industrial microbiologists. Its biochemical meaning relates to the generation of energy by the catabolism of organic compounds, whereas its meaning in industrial microbiology tends to be much broader.

The catabolism of sugars is an oxidative process which results in the production of reduced pyridine nucleotides which must be reoxidized for the process to continue. Under aerobic conditions, reoxidation of reduced pyridine nucleotide occurs by electron transfer, via the cytochrome system, with oxygen acting as the terminal electron acceptor. However, under anaerobic conditions, reduced pyridine nucleotide oxidation is coupled with the reduction of an organic compound, which is often a subsequent product of the catabolic pathway. In the case of the action of yeast on fruit or grain extracts, NADH is regenerated by the reduction of pyruvic acid to ethanol. Different microbial taxa are capable of reducing pyruvate to a wide range of end products, as illustrated in Fig. 1.1. Thus, the term fermentation has been used in a strict biochemical sense to mean an energy-generation process in which organic compounds act as both electron donors and terminal electron acceptors.

The production of alcohol by the action of yeast on malt or fruit extracts has been carried out on a large scale for very many years and was the first 'industrial' process for the production of a microbial metabolite. Thus, industrial microbiologists have extended the term fermentation to describe any process for the produc-

tion of product by the mass culture of a micro-organism. Brewing and the production of organic solvents may be described as fermentations in both senses of the word but the description of an aerobic process as a fermentation is obviously using the term in the broader, microbiological, context and it is in this sense that the term is used in this book.

THE RANGE OF FERMENTATION PROCESSES

There are five major groups of commercially important fermentations:

- (i) Those that produce microbial cells (or biomass) as the product.
- (ii) Those that produce microbial enzymes.
- (iii) Those that produce microbial metabolites.
- (iv) Those that produce recombinant products.
- (v) Those that modify a compound which is added to the fermentation — the transformation process.

The historical development of these processes will be considered in a later section of this chapter, but it is first necessary to include a brief description of the five groups.

Microbial biomass

The commercial production of microbial biomass may be divided into two major processes: the production of yeast to be used in the baking industry and the production of microbial cells to be used as human or animal food (single-cell protein). Bakers' yeast has been

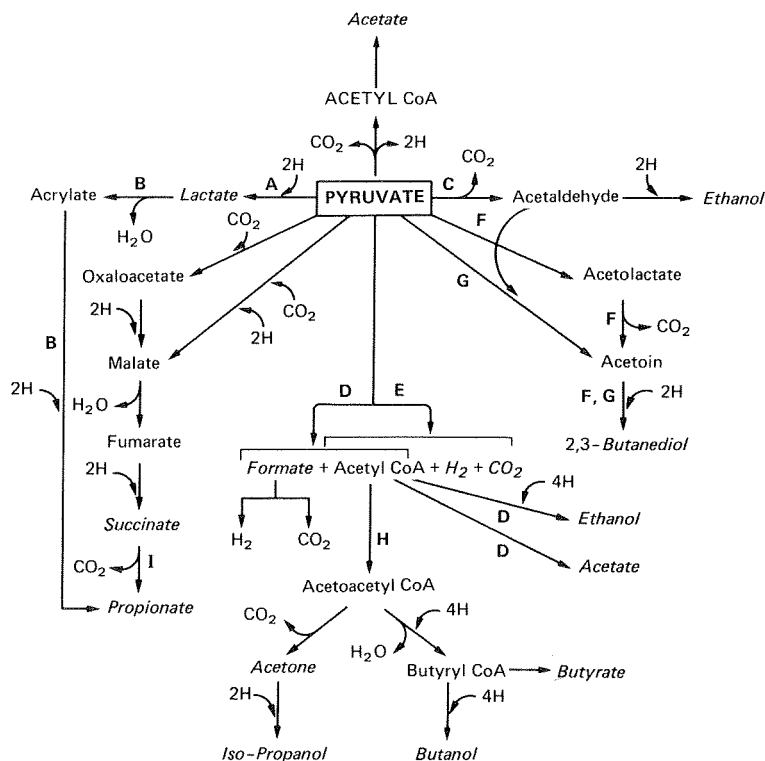


FIG. 1.1. Bacterial fermentation products of pyruvate. Pyruvate formed by the catabolism of glucose is further metabolized by pathways which are characteristic of particular organisms and which serve as a biochemical aid to identification. End products of fermentations are italicized (Dawes and Large, 1982).

A Lactic acid bacteria (*Streptococcus*, *Lactobacillus*)

B *Clostridium propionicum*

C Yeast, *Acetobacter*, *Zymomonas*, *Sarcina ventriculi*, *Erwinia amylovora*

E Clostridia

F *Klebsiella*

G Yeast

H Clostridia (butyric, butylic organisms)

I Propionic acid bacteria

produced on a large scale since the early 1900s and yeast was produced as human food in Germany during the First World War. However, it was not until the 1960s that the production of microbial biomass as a source of food protein was explored to any great depth. As a result of this work, reviewed briefly in Chapter 2, a few large-scale continuous processes for animal feed production were established in the 1970s. These processes were based on hydrocarbon feedstocks which could not compete against other high protein animal feeds, resulting in their closure in the late 1980s (Sharp, 1989). However, the demise of the animal feed biomass fermentations was balanced by ICI plc and Rank Hovis McDougal establishing a process for the production of fungal biomass for human food. This process was based on a more stable economic platform and appears to have a promising future.

Microbial enzymes

Enzymes have been produced commercially from plant, animal and microbial sources. However, microbial enzymes have the enormous advantage of being able to be produced in large quantities by established fermentation techniques. Also, it is infinitely easier to improve the productivity of a microbial system compared with a plant or animal one. Furthermore, the advent of recombinant DNA technology has enabled enzymes of animal origin to be synthesized by micro-organisms (see Chapter 3). The uses to which microbial enzymes have been put are summarized in Table 1.1, from which it may be seen that the majority of applications are in the food and related industries. Enzyme production is closely controlled in micro-organisms and in order to improve productivity these controls may

TABLE 1.1. *Commercial applications of enzymes* (Modified from Boing, 1982)

Industry	Application	Enzyme	Source
Baking and milling	Reduction of dough viscosity, acceleration of fermentation, increase in loaf volume, improvement of crumb softness and maintenance of freshness	Amylase	Fungal
	Improvement of dough texture, reduction of mixing time, increase in loaf volume	Protease	Fungal/bacterial
Brewing	Mashing	Amylase	Fungal/bacterial
	Chillproofing	Protease	Fungal/bacterial
Cereals	Improvement of fine filtration	β -Glucanase	Fungal/bacterial
	Precooked baby foods, breakfast foods	Amylase	Fungal
Chocolate and cocoa	Manufacture of syrups	Amylase	Fungal/bacterial
Coffee	Coffee bean fermentation	Pectinase	Fungal
	Preparation of coffee concentrates	Pectinase, hemicellulase	Fungal
Confectionery	Manufacture of soft centre candies	Invertase, pectinase	Fungal/bacterial
Corn syrup	Manufacture of high-maltose syrups	Amylase	Fungal
	Production of low D.E. syrups	Amylase	Bacterial
	Production of glucose from corn syrup	Amyloglycosidase	Fungal
	Manufacture of fructose syrups	Glucose isomerase	Bacterial
Dairy	Manufacture of protein hydrolysates	Protease	Fungal/bacterial
	Stabilization of evaporated milk	Protease	Fungal
	Production of whole milk concentrates, icecream and frozen desserts	Lactase	Yeast
	Curdling milk	Protease	Fungal/bacterial
Eggs, dried	Glucose removal	Glucose oxidase	Fungal
Fruit juices	Clarification	Pectinases	Fungal
	Oxygen removal	Glucose oxidase	Fungal
Laundry	Detergents	Protease, lipase	Bacterial
Leather	Dehairing, baiting	Protease	Fungal/bacterial
Meat	Tenderization	Protease	Fungal
Pharmaceutical	Digestive aids	Amylase, protease	Fungal
	Anti-blood clotting	Streptokinase	Bacterial
	Various clinical tests	Numerous	Fungal/bacterial
Photography	Recovery of silver from spent film	Protease	Bacterial
Protein hydrolysates	Manufacture	Proteases	Fungal/bacterial
Soft drinks	Stabilization	Glucose oxidase, catalase	Fungal
Textiles	Desizing of fabrics	Amylase	Bacterial
Vegetables	Preparation of purees and soups	Pectinase, amylase, cellulase	Fungal

have to be exploited or modified. Such control systems as induction may be exploited by including inducers in the medium (see Chapter 4), whereas repression control may be removed by mutation and recombination techniques. Also, the number of gene copies coding for the enzyme may be increased by recombinant DNA techniques. Aspects of strain improvement are discussed in Chapter 3.

Microbial metabolites

The growth of a microbial culture can be divided into a number of stages, as discussed in Chapter 2.

After the inoculation of a culture into a nutrient medium there is a period during which growth does not appear to occur; this period is referred to as the lag phase and may be considered as a time of adaptation. Following a period during which the growth rate of the cells gradually increases the cells grow at a constant, maximum rate and this period is known as the log, or exponential, phase. Eventually, growth ceases and the cells enter the so-called stationary phase. After a further period of time the viable cell number declines as the culture enters the death phase. As well as this kinetic description of growth, the behaviour of a culture may also be described according to the products

which it produces during the various stages of the growth curve. During the log phase of growth the products produced are essential to the growth of the cells and include amino acids, nucleotides, proteins, nucleic acids, lipids, carbohydrates, etc. These products are referred to as the primary products of metabolism and the phase in which they are produced (equivalent to the log, or exponential phase) as the trophophase (Bu'Lock *et al.*, 1965).

Many products of primary metabolism are of considerable economic importance and are being produced by fermentation, as illustrated in Table 1.2. The synthesis of primary metabolites by wild-type micro-organisms is such that their production is sufficient to meet the requirements of the organism. Thus, it is the task of the industrial microbiologist to modify the wild-type organism and to provide cultural conditions to improve the productivity of these compounds. This aspect is considered in Chapter 3.

During the deceleration and stationary phases some microbial cultures synthesize compounds which are not produced during the trophophase and which do not appear to have any obvious function in cell metabolism. These compounds are referred to as the secondary compounds of metabolism and the phase in which they are produced (equivalent to the stationary phase) as the idiophase (Bu'Lock *et al.*, 1965). It is important to realize that secondary metabolism may occur in continuous cultures at low growth rates and is a property of slow-growing, as well as non-growing, cells. When it is appreciated that micro-organisms grow at relatively low growth rates in their natural environments, it is tempting to suggest that it is the idiophase state that prevails in nature rather than the trophophase, which may be more of a property of micro-organisms in culture. The

inter-relationships between primary and secondary metabolism are illustrated in Fig. 1.2, from which it may be seen that secondary metabolites tend to be elaborated from the intermediates and products of primary metabolism. Although the primary biosynthetic routes illustrated in Fig. 1.2 are common to the vast majority of micro-organisms, each secondary product would be synthesized by only a very few different microbial species. Thus, Fig. 1.2 is a representation of the secondary metabolism exhibited by a very wide range of different micro-organisms. Also, not all micro-organisms undergo secondary metabolism — it is common amongst the filamentous bacteria and fungi and the sporing bacteria but it is not found, for example, in the Enterobacteriaceae. Thus, the taxonomic distribution of secondary metabolism is quite different from that of primary metabolism. It is important to appreciate that the classification of microbial products into primary and secondary metabolites is a convenient, but in some cases, artificial system. To quote Bushell (1988), the classification "should not be allowed to act as a conceptual straitjacket, forcing the reader to consider all products as either primary or secondary metabolites". It is sometimes difficult to categorize a product as primary or secondary and the kinetics of synthesis of certain compounds may change depending on the cultural conditions.

The physiological role of secondary metabolism in the producer cells has been the subject of considerable debate, but the importance of these metabolites to the fermentation industry is the effects they have on organisms other than those that produce them. Many secondary metabolites have antimicrobial activity, others are specific enzyme inhibitors, some are growth promoters and many have pharmacological properties. Thus, the products of secondary metabolism have formed the basis of a number of fermentation processes. As is the case for primary metabolites, wild-type micro-organisms tend to produce only low concentrations of secondary metabolites, their synthesis being controlled by induction, catabolite repression and feedback systems. The techniques which have been developed to improve secondary metabolite production are considered in Chapters 3 and 4.

Recombinant products

The advent of recombinant DNA technology has extended the range of potential fermentation products. Genes from higher organisms may be introduced into

TABLE 1.2. *Some primary products of microbial metabolism and their commercial significance*

Primary metabolite	Commercial significance
Ethanol	'Active ingredient' in alcoholic beverages Used as a motor-car fuel when blended with petroleum
Citric acid	Various uses in the food industry
Glutamic acid	Flavour enhancer
Lysine	Feed supplement
Nucleotides	Flavour enhancers
Phenylalanine	Precursor of aspartame, sweetener
Polysaccharides	Applications in the food industry Enhanced oil recovery
Vitamins	Feed supplements

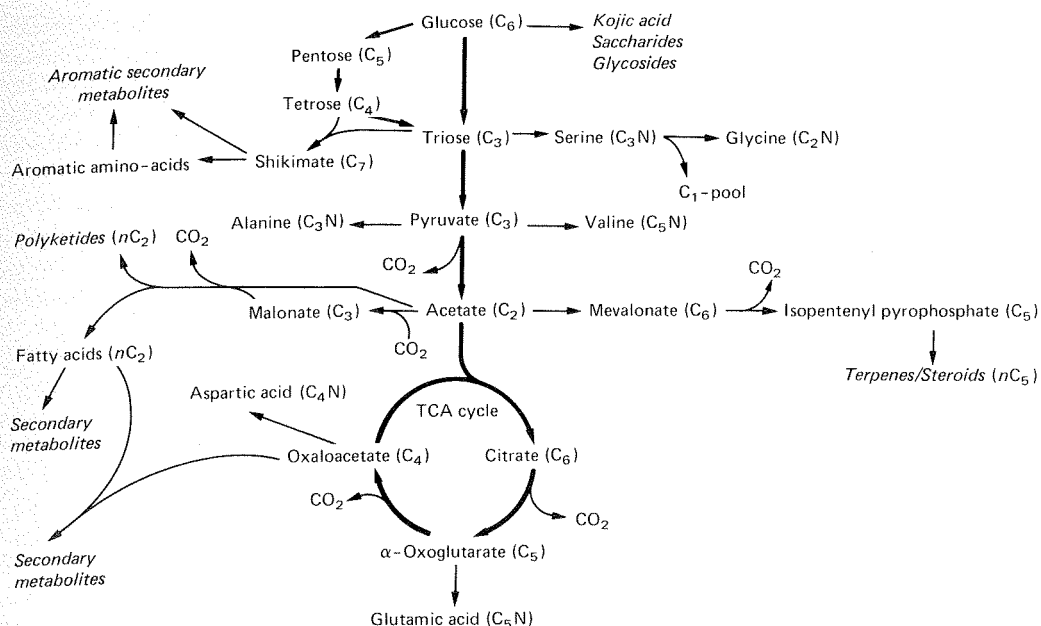


FIG. 1.2. The interrelationships between primary and secondary metabolism. Primary catabolic routes are shown in heavy lines and secondary products are italicized (Turner, 1971).

microbial cells such that the recipients are capable of synthesizing 'foreign' (or heterologous) proteins. A wide range of microbial cells have been used as hosts for such systems including *Escherichia coli*, *Saccharomyces cerevisiae* and filamentous fungi. Products produced by such genetically engineered organisms include interferon, insulin, human serum albumin, factors VIII and IX, epidermal growth factor, calf chymosin and bovine somatostatin. Important factors in the design of these processes include the secretion of the product, minimization of the degradation of the product and control of the onset of synthesis during the fermentation, as well as maximizing the expression of the foreign gene. These aspects are considered in more detail in Chapters 3 and 4.

Transformation processes

Microbial cells may be used to convert a compound into a structurally related, financially more valuable, compound. Because micro-organisms can behave as chiral catalysts with high positional specificity and stereospecificity, microbial processes are more specific than purely chemical ones and enable the addition, removal or modification of functional groups at specific sites on a complex molecule without the use of chemical protection. The reactions which may be catalysed

include dehydrogenation, oxidation, hydroxylation, dehydration and condensation, decarboxylation, amination, deamination and isomerization. Microbial processes have the additional advantage over chemical reagents of operating at relatively low temperatures and pressures without the requirement for potentially polluting heavy-metal catalysts. Although the production of vinegar is the most well-established microbial transformation process (conversion of ethanol to acetic acid) the majority of these processes involve the production of high-value compounds including steroids, antibiotics and prostaglandins.

The anomaly of the transformation fermentation process is that a large biomass has to be produced to catalyse a single reaction. Thus, many processes have been streamlined by immobilizing either the whole cells, or the isolated enzymes which catalyse the reactions, on an inert support. The immobilized cells or enzymes may then be considered as catalysts which may be reused many times.

THE CHRONOLOGICAL DEVELOPMENT OF THE FERMENTATION INDUSTRY

The chronological development of the fermentation industry may be represented as five overlapping stages

as illustrated in Table 1.3. The development of the industry prior to 1900 is represented by stage 1, where the products were confined to potable alcohol and vinegar. Although beer was first brewed by the ancient Egyptians, the first true large-scale breweries date from

the early 1700s when wooden vats of 1500 barrels capacity were introduced (Corran, 1975). Even some process control was attempted in these early breweries, as indicated by the recorded use of thermometers in 1757 and the development of primitive heat exchangers

TABLE 1.3. *The stages in the chronological development of the fermentation industry*

Stage	Main products	Vessels	Process control	Culture method	Quality control	Pilot plant facilities	Strain selection
1 Pre-1900	Alcohol	Wooden, up to 1500 barrels capacity Copper used in later breweries	Use of thermometer, hydrometer and heat exchangers	Batch	Virtually nil	Nil	Pure yeast cultures used at the Carlsberg brewery (1886)
	Vinegar	Barrels, shallow trays, trickle filters		Batch	Virtually nil	Nil	Fermentations inoculated with 'good' vinegar
2 1900-1940	Bakers' yeast glycerol, citric acid, lactic acid and acetone/ butanol	Steel vessels of up to 200 m ³ for acetone/butanol Air spargers used for bakers' yeast Mechanical stirring used in small vessels	pH electrodes with off-line control Temperature control	Batch and fed-batch systems	Virtually nil	Virtually nil	Pure cultures used
3 1940-date	Penicillin, streptomycin, other antibiotics, gibberelin, amino acids, nucleotides, transformations, enzymes	Mechanically aerated vessels, operated aseptically—true fermenters	Sterilizable pH and oxygen electrodes. Use of control loops which were later computerized	Batch and fed-batch common Continuous culture introduced for brewing and some primary metabolites	Very important	Becomes common	Mutation and selection programmes essential
4 1964-date	Single-cell protein using hydrocarbon and other feedstocks	Pressure cycle and pressure jet vessels developed to overcome gas and heat exchange problems	Use of computer linked control loops	Continuous culture with medium recycle	Very important	Very important	Genetic engineering of producer strains attempted
5 1979-date	Production of heterologous proteins by microbial and animal cells Monoclonal antibodies produced by animal cells	Fermenters developed in stages 3 and 4. Animal cell reactors developed	Control and sensors developed in stages 3 and 4	Batch, fed-batch or continuous Continuous perfusion developed for animal cell processes	Very important	Very important	Introduction of foreign genes into microbial and animal cell hosts. <i>In vitro</i> recombinant DNA techniques used in the improvement of stage 3 products

in 1801. By the mid-1800s the role of yeasts in alcoholic fermentation had been demonstrated independently by Cagniard-Latour, Schwann and Kützing but it was Pasteur who eventually convinced the scientific world of the obligatory role of these micro-organisms in the process. During the late 1800s Hansen started his pioneering work at the Carlsberg brewery and developed methods for isolating and propagating single yeast cells to produce pure cultures and established sophisticated techniques for the production of starter cultures. However, use of pure cultures did not spread to the British ale breweries and it is true to say that many of the small, traditional, ale-producing breweries still use mixed yeast cultures at the present time but, nevertheless, succeed in producing high quality products.

Vinegar was originally produced by leaving wine in shallow bowls or partially filled barrels where it was slowly oxidized to vinegar by the development of a natural flora. The appreciation of the importance of air in the process eventually led to the development of the 'generator' which consisted of a vessel packed with an inert material (such as coke, charcoal and various types of wood shavings) over which the wine or beer was allowed to trickle. The vinegar generator may be considered as the first 'aerobic' fermenter to be developed. By the late 1800s to early 1900s the initial medium was being pasteurized and inoculated with 10% good vinegar to make it acidic, and therefore resistant to contamination, as well as providing a good inoculum (Biolletti, 1921). Thus, by the beginning of the twentieth century the concepts of process control were well established in both the brewing and vinegar industries.

Between the years 1900 and 1940 the main new products were yeast biomass, glycerol, citric acid, lactic acid, acetone and butanol. Probably the most important advances during this period were the developments in the bakers' yeast and solvent fermentations. The production of bakers' yeast is an aerobic process and it was soon recognized that the rapid growth of yeast cells in a rich wort led to oxygen depletion in the medium which, in turn, resulted in ethanol production at the expense of biomass formation. The problem was minimized by restricting the initial wort concentration such that the growth of the cells was limited by the availability of the carbon source rather than oxygen. Subsequent growth of the culture was then controlled by adding further wort in small amounts. This technique is now called fed-batch culture and is widely used in the fermentation industry to avoid conditions of oxygen limitation. The aeration of these early yeast cultures was also improved by the introduction of air

through sparging tubes which could be steam cleaned (de Becze and Liebmann, 1944).

The development of the acetone-butanol fermentation during the First World War by the pioneering efforts of Weizmann led to the establishment of the first truly aseptic fermentation. All the processes discussed so far could be conducted with relatively little contamination provided that a good inoculum was used and reasonable standards of hygiene employed. However, the anaerobic butanol fermentation was susceptible to contamination in the early stages by aerobic bacteria, and by acid-producing anaerobic ones once anaerobic conditions had been established in the later stages of the process. The fermenters employed were vertical cylinders with hemispherical tops and bottoms constructed from mild steel. They could be steam sterilized under pressure and were constructed to minimize the possibility of contamination. Two-thousand-hectalitre fermenters were commissioned which presented the problems of inoculum development and the maintenance of aseptic conditions during the inoculation procedure. The techniques developed for the production of these organic solvents were a major advance in fermentation technology and paved the way for the successful introduction of aseptic aerobic processes in the 1940s.

The third stage of the development of the fermentation industry arose as a result of the wartime need to produce penicillin in submerged culture under aseptic conditions. The production of penicillin is an aerobic process which is very vulnerable to contamination. Thus, although the knowledge gained from the solvent fermentations was exceptionally valuable, the problems of sparging a culture with large volumes of sterile air and mixing a highly viscous broth had to be overcome. Also, unlike the solvent fermentations, penicillin was synthesized in very small quantities by the initial isolates and this resulted in the establishment of strain-improvement programmes which became a dominant feature of the industry in subsequent years. Process development was also aided by the introduction of pilot-plant facilities which enabled the testing of new techniques on a semi-production scale. The development of a large-scale extraction process for the recovery of penicillin was another major advance at this time. The technology established for penicillin fermentation provided the basis for the development of a wide range of new processes. This was probably the stage when the most significant changes in fermentation technology took place resulting in the establishment of many new processes over the period, including other antibiotics, vitamins, gibberellin, amino acids, enzymes and steroid

transformations. From the 1960s onwards microbial products were screened for activities other than simply antimicrobial properties and screens became more and more sophisticated. These screens have evolved into those operating today utilizing miniaturized culture systems, robotic automation and elegant assays.

In the early 1960s the decisions of several multi-national companies to investigate the production of microbial biomass as a source of feed protein led to a number of developments which may be regarded as the fourth stage in the progress of the industry. The largest mechanically stirred fermentation vessels developed during stage 3 were in the range 80,000 to 150,000 dm³. However, the relatively low selling price of microbial biomass necessitated its production in much larger quantities than other fermentation products in order for the process to be profitable. Also, hydrocarbons were considered as potential carbon sources which would result in increased oxygen demands and high heat outputs by these fermentations (see Chapters 4 and 9). These requirements led to the development of the pressure jet and pressure cycle fermenters which eliminated the need for mechanical stirring (see Chapter 7). Another feature of these potential processes was that they would have to be operated continuously if they were to be economic. At this time batch and fed-batch processes were common in the industry but the technique of growing an organism continuously by adding fresh medium to the vessel and removing culture fluid had been applied only to a very limited extent on a large scale. The brewers were also investigating the potential of continuous culture at this time, but its application in that industry was short-lived. Several companies persevered in the biomass field and a few processes came to fruition, of which the most long-lived was the ICI Pruteen animal feed process which utilized a continuous 3,000,000-dm³ pressure cycle fermenter for the culture of *Methylophilus methylotrophus* with methanol as carbon source (Smith, 1981; Sharp, 1989). The operation of an extremely large continuous fermenter for time periods in excess of 100 days presented a considerable aseptic operation problem, far greater than that faced by the antibiotic industry in the 1940s. The aseptic operation of fermenters of this type was achieved as a result of the high standards of fermenter construction, the continuous sterilization of feed streams and the utilization of computer systems to control the sterilization and operation cycles, thus minimizing the possibility of human error. However, although the Pruteen process was a technological triumph it became an economic failure because the

product was out-priced by soybean and fishmeal. Eventually, in 1989, the plant was demolished, marking the end of a short, but very exciting, era in the fermentation industry.

Whilst biomass is a very low-value, high-volume product, the fifth stage in the progress of the industry resulted in the establishment of very high-value, low-volume products. The developments in *in vitro* genetic manipulation, commonly known as genetic engineering, enabled the expression of human and mammalian genes in micro-organisms, thereby enabling the large scale production of human proteins which could then be used therapeutically. According to Dykes (1993) it was the small, venture-capital biotechnology companies that pioneered the development of heterologous proteins for therapeutic use. The established pharmaceutical companies used the new genetic engineering techniques to help in the discovery of natural products and in the rational design of drugs; for example, mammalian receptor proteins have been cloned and used in *in vitro* detection systems.

Table 3.8 (Chapter 3) lists the recombinant proteins licensed for therapeutic use. According to Dykes (1993), insulin and human growth hormone have been the two most successful products but other products have far greater potential. Erythropoietin and the myeloid colony stimulating factors (CSFs) control the production of blood cells by stimulating the proliferation, differentiation and activation of specific cell types. Erythropoietin has been used to treat renal-failure anaemia and may have application in the treatment of the platelet deficiency associated with cancer chemotherapy; it is expected to become the top-selling therapeutic protein by the mid-1990s with annual sales of around \$1200 million (Dykes, 1993). Granulocyte-colony stimulating factor (G-CSF), which is used during cancer chemotherapy, generated sales of over \$230 million in 1991 and, due to other uses, its sales could reach \$1000 million by 1996. A number of different growth factors are involved in wound healing and recombinant forms of these proteins would be expected to yield significant returns during the 1990s.

The commercial exploitation of recombinant proteins has necessitated the design of contained production facilities. Thus, these processes are drawing on the experience of vaccine fermentations where pathogenic organisms have been grown on relatively large scales. Also, recombinant proteins have been classified as biologicals, not as drugs, and thus come under the same regulatory authorities as do vaccines. The major difference between the approval of drugs and biologicals

is that the process for the production of a biological must be precisely specified and carried out in a facility that has been inspected and licensed by the regulatory authority, which is not the case for the production of drugs (antibiotics, for example) (Bader, 1992). Thus, any changes which a manufacturer wishes to incorporate into a licensed process must receive regulatory approval. For drugs, only major changes require approval prior to implementation. The result of these containment and regulatory requirements is that the cost of developing a recombinant protein process is extremely high. Buckland (1992) illustrated this point in his claim that "It now costs as much to build a 3000 dm³ scale facility for Biologics as for a 200,000 dm³ scale facility for an antibiotic. Also, even though titres are now reasonable for a recombinant protein (1 g dm⁻³), the cost of manufacture kg⁻¹ of bulk drug is about two orders of magnitude higher than that of an antibiotic at 10 g dm⁻³ titre". Also, the development time for a recombinant protein is considerably longer than that for an antibiotic. For example, Bader (1992) claimed that it is feasible for an antibiotic plant to begin production four years after the initiation of the plant design whereas seven years would be required before a recombinant protein could be produced.

The exploitation of genetic engineering approximately coincided with another major development in biotechnology which influenced the progress of the fermentation industry — the production of monoclonal antibodies. The availability of monoclonal antibodies opened the door to sophisticated analytical techniques and raised hopes for their use as therapeutic agents. Although the promise of therapeutic agents has yet to be realized (only one monoclonal antibody has been licensed for clinical use, OKT3, used in the treatment of acute renal allograft rejection (Webb, 1993)), their use as tools in biological research has increased exponentially. Thus, animal cell culture processes were established to produce monoclonals on a commercial scale. Subsequently, animal cells were also used as hosts for the production of some human proteins, especially where post-translational modification was essential for protein activity. Although these animal cell processes were based on microbial fermentation technology a number of novel problems had to be solved — animal cells are extremely fragile compared with microbial cells, the achievable cell density is very much less than in a microbial process and the media are very complex. These aspects are considered in Chapters 4 and 7.

The outstanding developments in recombinant fermentations (stage 5) have tended to overshadow the

progress which has been made in recent years in establishing new fermentations based on conventional microbial products (the continuing development of stage 4). However, the appreciation by the pharmaceutical industry that the activity of microbial metabolites extended well beyond antibacterials has resulted in a number of new microbial products reaching the marketplace in the late 1980s and early 1990s. Buckland (1992) listed four secondary metabolites which were launched in the 1980s: cyclosporin, an immunoregulant used to control rejection of transplanted organs; imipenem, a modified carbapenem which has the widest antimicrobial spectrum of any antibiotic; lovastatin, a drug used for reducing cholesterol levels and ivermectin, an anti-parasitic drug which has been used to prevent 'African River Blindness' as well as in veterinary practice. Buckland summarized these developments succinctly, "One of the best kept secrets (unintentionally kept as a secret) in the 1980s in Biochemical Engineering was that working on secondary metabolites was a fascinating, important and rewarding experience. Furthermore the four products listed added together have higher sales than all of the recombinant products added together". Thus, it is still relevant to heed Foster's warning (1949) "never underestimate the power of the microbe".

THE COMPONENT PARTS OF A FERMENTATION PROCESS

Regardless of the type of fermentation (with the possible exception of some transformation processes) an established process may be divided into six basic component parts:

- (i) The formulation of media to be used in culturing the process organism during the development of the inoculum and in the production fermenter.
- (ii) The sterilization of the medium, fermenters and ancillary equipment.
- (iii) The production of an active, pure culture in sufficient quantity to inoculate the production vessel.
- (iv) The growth of the organism in the production fermenter under optimum conditions for product formation.
- (v) The extraction of the product and its purification.
- (vi) The disposal of effluents produced by the process.

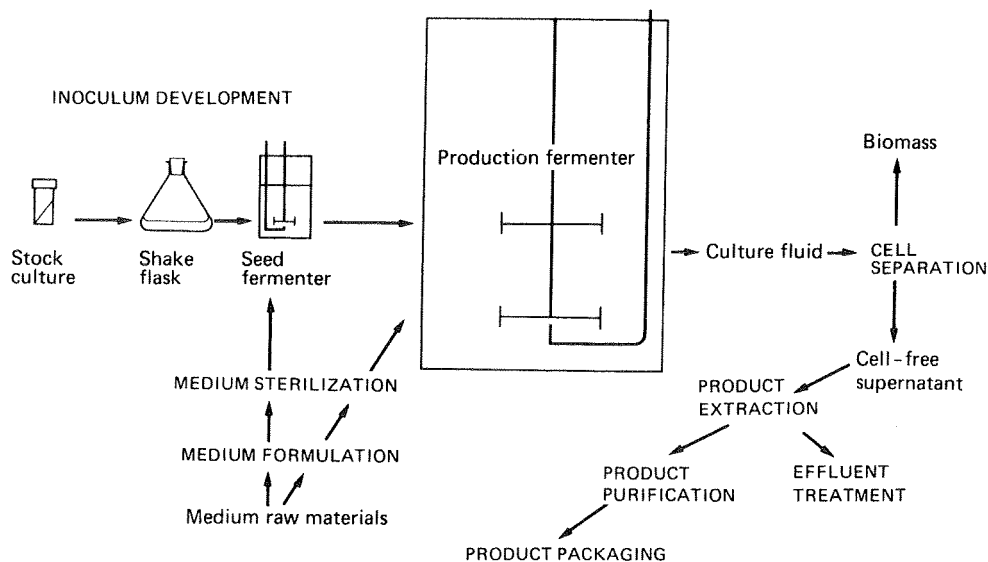


FIG. 1.3. A generalized schematic representation of a typical fermentation process.

The interrelationships between the six component parts are illustrated in Fig 1.3.

However, one must also visualize the research and development programme which is designed to gradually improve the overall efficiency of the fermentation. Before a fermentation process is established a producer organism has to be isolated, modified such that it produces the desired product in commercial quantities, its cultural requirements determined and the plant designed accordingly. Also, the extraction process has to be established. The development programme would involve the continual improvement of the process organism, the culture medium and the extraction process.

The subsequent chapters in this book consider the basic principles underlying the component parts of a fermentation. Chapter 2 considers growth, comprehension of which is crucial to understanding many aspects of the process, other than simply the growth of the organism in the production fermenter. The isolation and improvement of commercial strains is considered in Chapter 3 and the design of media in Chapter 4. The sterilization of the medium, fermenters and air is considered in Chapter 5 and the techniques for the development of inocula are discussed in Chapter 6. Chapters 7, 8 and 9 consider the fermenter as an environment for the culture of micro-organisms; Chapter 7 considers the design and construction of fermenters including contained systems and animal cell fermenters, Chapter 8 discusses the instrumentation involved in monitoring and maintaining a controlled environment in a fer-

menter, while the provision of oxygen to a culture is investigated in Chapter 9. The recovery of fermentation products is dealt with in Chapter 10 and the disposal of effluents of processes is covered in Chapter 11. Finally, the economics of fermentation processes are discussed in Chapter 12. Throughout the book examples are drawn from a very wide range of fermentations to illustrate the applications of the techniques being discussed but it has not been attempted to give detailed considerations of specific processes as this is well covered elsewhere, for example in the Biotechnology series edited by Rehm and Reed. We hope that the approach adopted in this book will give the reader an understanding of the basic principles underlying the techniques used for the large-scale production of microbial products.

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Microbial Growth Kinetics

AS OUTLINED in Chapter 1, fermentations may be carried out as batch, continuous and fed-batch processes. The mode of operation is, to a large extent, dictated by the type of product being produced. This chapter will consider the kinetics and applications of batch, continuous and fed-batch processes.

BATCH CULTURE

Batch culture is a closed culture system which contains an initial, limited amount of nutrient. The inoculated culture will pass through a number of phases, as illustrated in Fig. 2.1. After inoculation there is a period during which it appears that no growth takes place; this period is referred to as the lag phase and may be considered as a time of adaptation. In a commercial process the length of the lag phase should be reduced as much as possible and this may be achieved by using a suitable inoculum, as described in Chapter 6. Following a period during which the growth rate of the cells gradually increases, the cells grow at a constant, maximum, rate and this period is known as the log, or exponential, phase. The exponential phase may be described by the equation:

$$dx/dt = \mu x \quad (2.1)$$

where x is the concentration of microbial biomass,
 t is time, in hours

and μ is the specific growth rate, in hours^{-1} .

On integration equation (2.1) gives:

$$x_t = x_0 e^{\mu t} \quad (2.2)$$

where x_0 is the original biomass concentration,
 x_t is the biomass concentration after the time interval, t hours,
 e is the base of the natural logarithm.

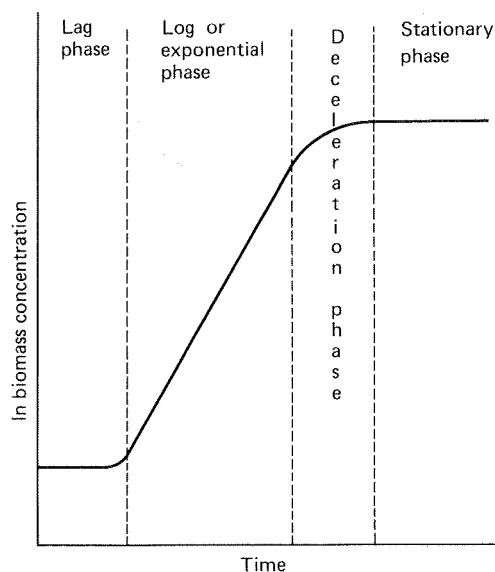


FIG. 2.1. Growth of a typical microbial culture in batch conditions.

On taking natural logarithms, equation (2.2) becomes:

$$\ln x_t = \ln x_0 + \mu t. \quad (2.3)$$

Thus, a plot of the natural logarithm of biomass concentration against time should yield a straight line, the slope of which would equal μ . During the exponential phase nutrients are in excess and the organism is growing at its maximum specific growth rate, $\mu_{\max}$, for the prevailing conditions. Typical values of $\mu_{\max}$ for a range of micro-organisms are given in Table 2.1.

It is easy to visualize the exponential growth of single celled organisms which replicate by binary fission. Indeed, animal and plant cells in suspension culture will behave very similarly to unicellular micro-organisms (Griffiths, 1986; Petersen and Alfermann, 1993). However, it is more difficult to appreciate that

TABLE 2.1. Some representative values of $\mu_{\max}$ (obtained under the conditions specified in the original reference) for a range of organisms

Organism	$\mu_{\max}$ (h^{-1})	Reference
<i>Vibrio natriegens</i>	4.24	Eagon (1961)
<i>Methylobomonas methanolytica</i>	0.53	Dostalek <i>et al.</i> (1972)
<i>Aspergillus nidulans</i>	0.36	Trinci (1969)
<i>Penicillium chrysogenum</i>	0.12	Trinci (1969)
<i>Fusarium graminearum</i> Schwabe	0.28	Trinci (1992)
Plant cells in suspension culture	0.01–0.046	Petersen and Alfermann (1993)
Animal cells	0.01–0.05	Lavery (1990)

mycelial organisms which show apical growth also grow exponentially. Plomley (1959) was the first to suggest that filamentous fungi have a 'growth unit' which is replicated at a constant rate and is composed of the apex of the hypha and a short length of supporting hypha. Trinci (1974) demonstrated that the total hyphal length of a mycelium and the number of tips increased exponentially at approximately the same rate. Thus, when the volume of the hyphal growth unit exceeds a critical volume a new branch, and hence, a new growing point, is initiated. This is equivalent to the division of a single cell when the cell reaches a critical volume. Hence, the rate of increase in hyphal mass, total length and number of tips is dictated by the specific growth rate and:

$$dx/dt = \mu x,$$

$$dH/dt = \mu H,$$

$$dA/dt = \mu A$$

where H is total hyphal length and A is the number of growing tips.

In submerged culture (shake flask or fermenter) a mycelial organism may grow as dispersed hyphal fragments or as pellets (see also Chapters 6 and 9). The growth of pellets will be exponential until the density of the pellet results in diffusion limitation. Under such limitation the central biomass of the pellet will not receive a supply of nutrients, nor will potentially toxic products diffuse out. Thus, the growth of the pellet proceeds from the outer shell of biomass which is the actively growing zone and was described by Pirt (1975) as:

$$M^{1/3} = kt + M_0^{1/3}$$

where M_0 and M are the mycelium mass at time 0 and t , respectively. Thus, a plot of the cube root of mycelial mass against time will give a straight line, the slope of which equals k .

It is possible for new pellets to be generated by the fragmentation of old pellets and, thus, the behaviour of a pelleted culture may be intermediate between exponential and cube root growth.

Whether the organism is unicellular or mycelial the foregoing equations predict that growth will continue indefinitely. However, growth results in the consumption of nutrients and the excretion of microbial products; events which influence the growth of the organism. Thus, after a certain time the growth rate of the culture decreases until growth ceases. The cessation of growth may be due to the depletion of some essential nutrient in the medium (substrate limitation), the accumulation of some autotoxic product of the organism in the medium (toxin limitation) or a combination of the two.

The nature of the limitation of growth may be explored by growing the organism in the presence of a range of substrate concentrations and plotting the biomass concentration at stationary phase against the initial substrate concentration, as shown in Fig. 2.2. From Fig. 2.2 it may be seen that over the zone A to B an increase in initial substrate concentration gives a proportional increase in the biomass produced at stationary phase. The situation may be described by the equation:

$$x = Y(S_R - s) \quad (2.4)$$

where x is the concentration of biomass produced, Y is the yield factor ($\text{g biomass produced g}^{-1}$ substrate consumed), S_R is the initial substrate concentration, and s is the residual substrate concentration.

Over the zone A to B in Fig. 2.2, s equals zero at the point of cessation of growth. Thus, equation (2.4) may be used to predict the biomass which may be produced

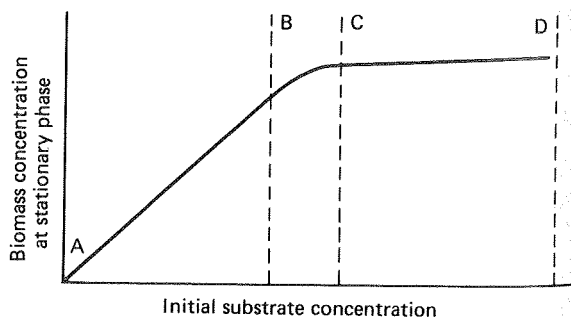


FIG. 2.2. The effect of initial substrate concentration on the biomass concentration at the onset of stationary phase, in batch culture.

from a certain amount of substrate. Over the zone C to D an increase in the initial substrate concentration does not give a proportional increase in biomass. This may be due to either the exhaustion of another substrate or the accumulation of toxic products. Over the zone B to C the utilization of the substrate is deleteriously affected by either the accumulating toxins or the availability of another substrate.

The yield factor (Y) is a measure of the efficiency of conversion of any one substrate into biomass and it can be used to predict the substrate concentration required to produce a certain biomass concentration. However, it is important to appreciate that Y is not a constant — it will vary according to growth rate, pH, temperature, the limiting substrate and the concentration of the substrates in excess.

The decrease in growth rate and the cessation of growth, due to the depletion of substrate, may be described by the relationship between μ and the residual growth-limiting substrate, represented in equation (2.5) and in Fig. 2.3 (Monod, 1942):

$$\mu = \mu_{\max} s / (K_s + s) \quad (2.5)$$

where s is the residual substrate concentration, K_s is the substrate utilization constant, numerically equal to substrate concentration when μ is half $\mu_{\max}$ and is a measure of the affinity of the organism for its substrate.

The zone A to B in Fig. 2.3 is equivalent to the exponential phase in batch culture where substrate concentration is in excess and growth is at $\mu_{\max}$. The zone C to A in Fig. 2.3 is equivalent to the deceleration phase of batch culture where the growth of the organism has resulted in the depletion of substrate to a

growth-limiting concentration which will not support $\mu_{\max}$. If the organism has a very high affinity for the limiting substrate (a low K_s value) the growth rate will not be affected until the substrate concentration has declined to a very low level. Thus, the deceleration phase for such a culture would be short. However, if the organism has a low affinity for the substrate (a high K_s value) the growth rate will be deleteriously affected at a relatively high substrate concentration. Thus, the deceleration phase for such a culture would be relatively long. Typical values of K_s for a range of organisms and substrates are shown in Table 2.2, from which it may be seen that such values are usually very small and the affinity for substrate is high. It will be appreciated that the biomass concentration at the end of the exponential phase is at its highest and, thus, the decline in substrate concentration will be very rapid so that the time period during which the substrate concentration is close to K_s is very short.

The stationary phase in batch culture is that point where the growth rate has declined to zero. However, as Bull (1974) pointed out, the stationary phase is a misnomer in terms of the physiology of the organism, as the population is still metabolically active during this phase and may produce products called secondary metabolites, which are not produced during the exponential phase. Bull suggested that this phase be termed the maximum population phase. The metabolic activity of the stationary phase has been recognized in the physiological descriptions of microbial growth presented by Borrow *et al.* (1961) and Bu'Lock *et al.* (1965). Borrow *et al.* investigated the biosynthesis of gibberellic acid by *Gibberella fujikuroi* and divided the growth of the organism into several phases:

- (i) The balanced phase; equivalent to the early to middle exponential phase.
- (ii) The storage phase; equivalent to the late exponential phase where the increase in mass is due to the accumulation of lipid and carbohydrate.

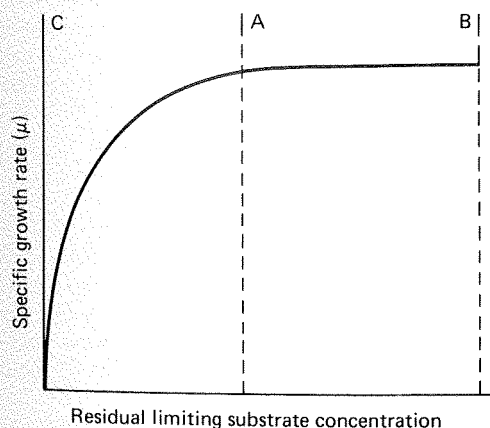


FIG. 2.3. The effect of residual limiting substrate concentration on the specific growth rate of a hypothetical bacterium.

TABLE 2.2. Some representative values of K_s for a range of micro-organisms and substrates

Organism	Substrate	K_s (mg dm ⁻³)	References
<i>Escherichia coli</i>	Glucose	6.8×10^{-2}	Shehata and Marr (1971)
<i>Saccharomyces cerevisiae</i>	Glucose	25.0	Pirt and Kurowski (1970)
<i>Pseudomonas</i> sp.	Methanol	0.7	Harrison (1973)

- (iii) The maintenance phase; equivalent to the stationary phase.

Gibberellic acid (a secondary metabolite) was synthesized only towards the end of the storage phase and during the maintenance phase. As discussed in Chapter 1, Bu'Lock *et al.* (1965) coined the terms trophophase, to refer to the exponential phase, and idiophase to refer to the stationary phase where secondary metabolites are produced. The idiophase was depicted as the period subsequent to the exponential phase in which secondary metabolites were synthesized. However, it is now obvious that the culture conditions may be manipulated to induce secondary metabolism during logarithmic growth, for example by the use of carbon sources which support a reduced maximum growth rate (see Chapter 4).

Pirt (1975) has discussed the kinetics of product formation by microbial cultures in terms of growth-linked products and non-growth-linked products. Growth-linked may be considered equivalent to primary metabolites which are synthesized by growing cells and non-growth-linked may be considered equivalent to secondary metabolites. The formation of a growth-linked product may be described by the equation:

$$dp/dt = q_p x \quad (2.6)$$

where p is the concentration of product and q_p is the specific rate of product formation (mg product g^{-1} biomass h^{-1})

Also, product formation is related to biomass production by the equation:

$$dp/dx = Y_{p/x} \quad (2.7)$$

where $Y_{p/x}$ is the yield of product in terms of biomass (g product g^{-1} biomass)

Multiply equation (2.7) by dx/dt , then:

$$dx/dt \cdot dp/dx = Y_{p/x} \cdot dx/dt$$

and $dp/dt = Y_{p/x} \cdot dx/dt$.

But $dx/dt = \mu x$ and therefore:

$$dp/dt = Y_{p/x} \cdot \mu x$$

and $dp/dt = q_p \cdot x$

and therefore:

$$\begin{aligned} q_p \cdot x &= Y_{p/x} \cdot \mu x, \\ q_p &= Y_{p/x} \cdot \mu. \end{aligned} \quad (2.8)$$

From equation (2.8) it may be seen that when product

formation is growth associated the specific rate of product formation increases with specific growth rate. Thus, productivity in batch culture will be greatest at μ_{max} and improved product output will be achieved by increasing both μ and biomass concentration. Non-growth linked product formation is related to biomass concentration and, thus, increased productivity in batch culture should be associated with an increase in biomass. However, it should be remembered that non-growth related secondary metabolites are produced only under certain physiological conditions — primarily under limitation of a particular substrate so that the biomass must be in the correct 'physiological state' before production can be achieved. The elucidation of the environmental conditions which create the correct 'physiological state' is extremely difficult in batch culture and this aspect is developed in a later section.

Thus, batch fermentation may be used to produce biomass, primary metabolites and secondary metabolites. For biomass production, cultural conditions supporting the fastest growth rate and maximum cell population would be used; for primary metabolite production conditions to extend the exponential phase accompanied by product excretion and for secondary metabolite production, conditions giving a short exponential phase and an extended production phase, or conditions giving a decreased growth rate in the log phase resulting in earlier secondary metabolite formation.

CONTINUOUS CULTURE

Exponential growth in batch culture may be prolonged by the addition of fresh medium to the vessel. Provided that the medium has been designed such that growth is substrate limited (i.e. by some component of the medium), and not toxin limited, exponential growth will proceed until the additional substrate is exhausted. This exercise may be repeated until the vessel is full. However, if an overflow device were fitted to the fermenter such that the added medium displaced an equal volume of culture from the vessel then continuous production of cells could be achieved. If medium is fed continuously to such a culture at a suitable rate, a steady state is achieved eventually, that is, formation of new biomass by the culture is balanced by the loss of cells from the vessel. The flow of medium into the vessel is related to the volume of the vessel by the term dilution rate, D , defined as:

$$D = F/V \quad (2.9)$$

where F is the flow rate ($dm^3 h^{-1}$)